Inducing Causal Meta-Knowledge From Virtual Domain: Causal Meta-Generalization for Hyperspectral Domain Generalization

Haoyu Wang, *Member, IEEE*, Xiaomin Liu, *Member, IEEE*, Zhenzhuang Qiao, and Hanqing Tao

***Abstract*— Cross-domain hyperspectral image (HSI) classification can improve the model's classification performance in the target domain by utilizing the rich knowledge from the source domain. However, existing cross-domain HSI classification methods mostly belong to transductive learning, which is difficult to apply to domain generalization tasks where the target domain is unseen during model learning. Inspired by human causal reasoning and knowledge induction mechanisms, this article develops an inductive learning-based framework for hyperspectral domain generalization: Causal meta-generalization. By simulating domain generalization scenarios, the framework helps the model induct domain-invariant causal meta-knowledge, thereby ensuring its strong generalization ability to unseen target domains. Specifically, we first propose a bottleneck variational auto-encoder (B-VAE) based on a forward–reverse information bottleneck, decoupling the domain distribution and class distribution of HSIs. By perturbing the domain distribution to generate virtual domains, we simulate potential domain distribution changes in the real world, providing a data basis for the induction of causal meta-knowledge. Second, in the process of simulating domain generalization scenarios, we establish a dual-layer optimization mechanism (DLOM) based on invariant-generalization risk minimization. In the inner layer optimization, by minimizing the model's invariant causal effect loss (ICEL) in the virtual source domains, we guide the model to learn domain-invariant causal meta-knowledge. In the outer optimization, by minimizing the model's generalization risk in the unseen virtual target domain, we enhance the applicability of causal meta-knowledge in domain generalization tasks. This proposed method has potential applications in remote sensing signal-processing tasks, such as the recognition of crop pests and diseases and the identification of minerals. The code can be accessed at https://github.com/wzr78998/CMG.**

***Index Terms*— Classification, cross-domain, domain generalization, hyperspectral image (HSI), meta-knowledge.**

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I. Introduction

HYPERSPECTRAL image (HSI) classification methods enable accurate identification of ground objects by deeply analyzing the rich spectral information contained in HSIs, demonstrating significant research value and application potential in fields such as agriculture, ecology, and environmental monitoring [1], [2], [3]. In recent years, HSI classification methods based on deep learning have garnered widespread attention from researchers [4], [5], [6]. However, the high performance of existing deep learning methods heavily relies on precise and abundant labeled data. In practical applications, obtaining labeled HSIs often involves high annotation costs [7]. Therefore, the inherent conflict between the reliance on labeled data in current methods and the high annotation costs in real-world scenarios limits the widespread deployment of HSI classification methods in practical applications.

To address this issue, researchers have proposed a series of cross-domain methods aimed at leveraging existing source-domain data with abundant annotations to train models, thereby reducing their dependence on labeled data from the target domain [8], [9]. The mainstream cross-domain methods primarily include distribution alignment-based methods and fine-tuning-based methods. Distribution alignment-based methods [10], [11], [12], [13] seek to model the data distributions of both the source domain and the unlabeled target domain using source-domain data, facilitating cross-domain knowledge transfer by minimizing the distribution discrepancy between the two. However, these methods often require the simultaneous participation of both source- and target-domain data during training, which may impose significant memory burdens, limiting their applicability in low-resource scenarios such as onboard satellite devices [14]. In contrast, fine-tuning-based methods [15], [16] aim to fine-tune the model trained on the source domain using target-domain data to enhance its adaptability to the target domain. However, models trained through this method are typically customized for a specific target domain, requiring retraining whenever a new target domain is encountered. With the advancement of remote sensing technology and the rapid growth of HSI data, customizing and retraining a model for each new target domain becomes not only time-consuming and labor-intensive but also potentially impractical.

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Domain generalization [14] has gained significant attention as it aims to inductively learn cross-domain shared knowledge that can be applied to unseen target domains, using only source-domain data. The key challenge of domain generalization lies in the fact that the target domain remains unknown during training. Current mainstream domain generalization methods include distribution expansion-based methods [14] and semantics-based methods [17], [18]. Distribution expansion-based methods aim to intervene in domain-specific information by using techniques such as data augmentation [19] and domain randomization [20] to expand the source-domain data distribution. The goal is for the expanded source-domain distribution to cover potential target-domain distributions. However, these methods face two main challenges. First, during the manipulation of source-domain data, they often make overly simplistic assumptions about domain-specific information, lacking mechanisms to decouple domain-specific information from domain-invariant information. Since domain-specific and domain-invariant information in data are often tightly coupled, interference with domain-specific information can harm the domain-invariant information, leading to suboptimal performance. Second, while expanding the source-domain distribution to cover potential target-domain distributions may yield good performance when the source and target domains have minimal differences, it becomes impractical when the domains differ significantly. Moreover, the optimization objective of these methods is to minimize the empirical risk on the expanded domain, but when the overlap between the target domain and the expanded source domain is low, this objective fails to ensure robust generalization on unseen domains. Semantics-based methods aim to model domain-invariant distributions by leveraging cross-domain shared semantic information to mitigate the cross-domain knowledge transfer issues caused by domain distribution discrepancies. However, these methods also have two limitations. First, their performance depends on high-quality textual annotations, which introduces additional labeling costs. Second, these methods often force sample feature distributions to approximate prior semantic distributions, resulting in the excessive transfer of domain-specific information, which can harm the model's domain generalization performance.

Moreover, it is important to note that most existing domain generalization methods focus primarily on establishing statistical associations between data and labels during training, neglecting the learning of cross-domain invariant causal relationships. This limits their potential for further performance improvement. For instance, when HSIs are collected during the day, the spectral signatures of objects often exhibit higher illumination intensity, leading models to establish statistical associations between daytime light intensity and class labels. However, when the target-domain HSIs are collected under low-light conditions, the statistical associations formed in the source domain may not hold, significantly limiting the model's generalization performance in the target domain. By incorporating a causal knowledge induction mechanism during model training, it is possible to uncover cross-domain invariant causal relationships between data and class labels, while cutting off spurious statistical associations, such as the one between light intensity and class labels in the aforementioned example. This would enhance the model's ability to generalize to unseen target domains.

To address the aforementioned issues, we propose the following solution: First, a decoupling mechanism for class and domain distributions. We aim to establish a mechanism that decouples class and domain distributions, with domain distribution encoding domain-specific information and class distribution encoding domain-invariant information. By separating domain-specific and -invariant information, we prevent the manipulation of source-domain data from harming domain-invariant information and mitigate the excessive transfer of domain-specific information. Second, overcoming the limitations of distribution expansion-based domain generalization. Instead of attempting to expand the source-domain distribution to cover potential target-domain distributions, our model is designed to "learn domain generalization." We achieve this through a meta-learning-based knowledge induction mechanism that perturbs domain-specific information to generate virtual domains. Based on these virtual domains, we construct domain generalization meta-tasks, allowing the model to learn how to capture cross-domain invariant general knowledge through domain generalization scenario simulations. Third, causal intervention mechanism based on front-door adjustment. We propose to establish a causal intervention mechanism using a front-door adjustment to sever spurious statistical associations between noncausal factors and labels. This enables precise capture of the true causal effect between data and labels, further enhancing the model's ability to generalize to unseen domains.

Specifically, we first propose a bottleneck variational autoencoder (B-VAE) that leverages the positive and inverse information bottleneck to decouple the tightly coupled domain and class distributions in HSIs. By perturbing the domain distribution, we simulate potential domain distribution changes in real-world HSI scenarios. With VAE's powerful data generation capability, we generate virtual domains, providing a data foundation for causal meta-knowledge induction. Second, we propose a dual-level optimization-based causal meta-knowledge induction mechanism. In the inner optimization loop, we construct meta-training tasks containing multiple virtual source domains and combine invariant risk minimization with a causal intervention mechanism by designing a causal effect loss for the meta-training tasks. By encouraging invariant causal effects among samples of the same class across different virtual domains, the model is guided to induce cross-domain transferable causal meta-knowledge. In the outer optimization loop, we construct meta-validation tasks that include unseen virtual target domains to simulate domain generalization scenarios from visible source domains to unseen target domains. By minimizing the model's generalization risk on the unseen target domains, we ensure the strong generalization capability of the causal meta-knowledge in unseen domains.

The contributions of this article are summarized as follows.

1) The causal meta-generalization framework is proposed, which constructs domain generalization meta-tasks

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through virtual domain generation and uses the knowledge induction mechanism of meta-learning to induce domain-invariant causal meta-knowledge, providing a new paradigm for hyperspectral domain generalization.

2) The distribution decoupling mechanism based on the positive-inverse information bottleneck is established, successfully decoupling class distribution from domain distribution and effectively separating domain-specific from domain-invariant information. This mechanism prevents damage to domain-invariant information during the virtual domain generation process and effectively mitigates the excessive transfer of domain-specific information.
3) The invariant causal effect loss (ICEL) based on causal structural models and causal intervention mechanisms is proposed, where the invariant causal effects between the same class of HSIs and labels across different virtual domains are constrained, enabling the accurate capture of domain-invariant causal relationships between HSIs and labels.
4) The limitations of existing methods that rely on empirical risk minimization optimization mechanisms have been overcome by proposing a dual-level optimization mechanism based on invariant-generalization risk minimization. This method balances the model's ability to induct domain-invariant causal knowledge and its cross-domain generalization capability for unseen target domain-specific tasks.

II. Related Work

A. Domain Adaptation

Remote sensing images offer precise spatial structure and feature information about ground objects. However, the labeling process is often costly, and target scenes (target domains) typically lack the sufficient annotated data required by models for effective training. To address this issue, researchers have explored domain adaptation methods [21], [22], which aim to leverage the knowledge from richly labeled available scenes (source domain) to improve the model's performance in the target domain. The mainstream domain adaptation methods currently include distribution alignment-based methods and self-supervised learning-based methods.

Distribution alignment-based methods aim to align the feature distributions between the source and target domains, reducing the distribution discrepancy and thus improving the model's performance in the target domain. Othman et al. [10] explored the potential of pretraining techniques and cross-domain distribution alignment in remote sensing red–green–blue (RGB) image classification tasks, utilizing pretrained convolutional neural networks to obtain initial feature representations for both domains. They then minimized the maximum mean discrepancy between the two domains to facilitate knowledge transfer across different domains. Gui et al. [12] focused on synthetic aperture radar (SAR) image classification tasks and designed a general feature paradigm for unsupervised cross-domain SAR image classification. They reduced the data distribution discrepancy between the source and target domains through polarization basis extraction, Wishart clustering, histogram statistics, and dimensionality reduction. Lu et al. [13] highlighted the challenge of finding a source domain that fully covers all target-domain classes, and in RGB remote sensing image classification tasks, addressed this by using multiple complementary source domains to cover the target-domain classes and mapping both domains into a common feature space to mitigate distribution shifts. Ma et al. [23] employed global attention in RGB remote sensing image segmentation tasks to capture shared knowledge across domains, reducing domain differences, and further enhancing cross-domain distribution alignment through an adversarial process between feature extractors and domain discriminators. Cheng et al. [11] introduced a novel causal learning perspective by establishing a causal intervention mechanism based on adversarial domain adaptation, which cuts off spurious statistical associations between data and labels to capture invariant causal factors across domains, thereby enhancing the model's performance in HSI classification tasks in the target domain, although it is important to note that most of these methods overlook the presence of domain-specific information. It is important to note that most of the above methods overlook the existence of domain-specific information. During distribution alignment, the forced transfer of domain-specific information can induce negative transfer, undermining the model's ability to transfer knowledge across domains.

Self-supervised learning-based methods facilitate cross-domain knowledge transfer by establishing self-supervised tasks that help models learn domain-invariant knowledge. Liu et al. [24] designed an RGB remote sensing image object detection framework by perturbing the color and style information of samples, creating a perturbation domain alignment task. This task constrained the detector to maintain consistent learning of domain-invariant knowledge across both the perturbation and target domains. Song et al. [25] developed a multitask unsupervised domain adaptation framework based on synthetic RGB remote sensing images. Through self-training, the framework learned domain-invariant knowledge, enabling knowledge transfer from synthetic data to real target data. Li et al. [22] proposed a contrastive learning-based framework for cross-domain HSI classification tasks. By constructing self-supervised learning tasks, the framework enforced similar data distributions between samples from different domains, enhancing the model's ability to transfer knowledge across domains. Zhao et al. [26] designed a self-training framework for heterogeneous SAR image classification, using clustering techniques to divide samples from both domains into class clusters to generate supervision signals. By minimizing the similarity entropy between domain samples and class prototypes, the framework helped the model learn domain-invariant knowledge. It should be noted that the above methods generally rely on manually designed self-supervised learning tasks to assist models in learning domain-invariant knowledge. When these tasks are not closely related to the downstream tasks, they lead to suboptimal model performance.

To address the aforementioned issues, in this article, we employ a B-VAE to decouple domain distribution and

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class distribution, enabling the separation of domain-invariant and -specific information. This method helps prevent negative transfer induced by the forced migration of domain-specific information. Furthermore, to strengthen the relevance between the meta-tasks and domain generalization tasks in HSI classification, we have customized domain generalization meta-tasks specifically for the HSI classification problem. By leveraging the strong inductive capabilities of meta-learning, we aim to capture cross-task shared knowledge, thereby improving the model's adaptability and generalization performance in HSI domain generalization tasks.

B. Domain Generalization

Domain generalization aims to train a model using only source-domain data, enabling the model to generalize to unseen target domains with different data distributions [27]. Although this ability is not difficult for humans, it remains a highly challenging problem for current deep learning models. Enhancing the domain generalization performance of models is still an open issue. The mainstream domain generalization methods include distribution expansion-based methods and semantics-based methods.

Key techniques involved in distribution expansion-based methods include data augmentation [19] and domain randomization [20]. These methods aim to expand the source-domain distribution to potentially cover the unseen target-domain distribution, thereby reducing the risk of the model when handling unseen domains. Iizuka et al. [20] explored domain randomization [28] solution for RGB remote sensing image semantic segmentation tasks by randomizing the style and appearance of the source domain to simulate potential variations in the target-domain distribution. They employed consistency regularization to enable the model to stably learn the same semantics from different images. Such methods primarily mitigate the impact of style and appearance differences on cross-domain knowledge transfer, but they struggle to capture the complex attribute differences between different domains, limiting their domain generalization performance. In addition, since style information can be tightly coupled with critical semantic information, intervening with style information might hinder the model's ability to learn key semantic features. Zhang et al. [14] proposed a single-source domain expansion network (SDEnet) for HSI classification, which generates virtual domains through data augmentation of source-domain data, thereby expanding the source-domain data distribution in hopes of covering the unseen target-domain distribution. However, it should be noted that in domain generalization tasks, the target-domain data distribution often possesses unpredictable complexity. Therefore, attempting to fully cover all possible target-domain distributions is difficult and often impractical. Moreover, the optimization objective of these methods is to minimize the empirical risk on the expanded domain, but when the overlap between the target domain and the expanded domain is low, this optimization objective may fail to ensure robust generalization performance on unseen domains.

Semantics-based methods [17], [18] leverage cross-domain shared semantic information to model domain-invariant distributions, aiming to alleviate the problem of cross-domain knowledge transfer being hindered by domain distribution discrepancies. Singha et al. [29] explored the potential of vision–language models in domain generalization for RGB remote sensing image classification tasks, using carefully designed guiding texts as prior knowledge to enhance the model's cross-domain generalization performance. Zhang et al. [35] attempted to learn domain-invariant feature representations from cross-domain shared language priors in HSI classification tasks. However, since HSI data is not pretrained, its spatial–spectral features cannot be directly aligned with language features. To address this, Dang et al. [17] proposed a cross-attention mechanism called "language and vision interactor," which enhances the exchange of information between the two modalities to facilitate cross-modal alignment. It should be noted that, on the one hand, the performance of such methods relies on high-quality text annotations, which introduces additional labeling costs. On the other hand, these methods often force the sample feature distributions to approximate the prior semantic distributions, leading to excessive transfer of domain-specific information, ultimately harming the model's cross-domain generalization performance.

To address these issues, we employ domain generalization scenario simulation to induct shared knowledge across tasks from meta-tasks and design a dual-level optimization mechanism based on invariant-generalization risk minimization. This enables the model to "learn domain generalization," meaning it can learn domain-invariant knowledge with strong generalization capabilities from different domains. In addition, we decouple domain distribution and class distribution, where the former encodes domain-specific information and the latter encodes domain-invariant information. During the model's learning process, we focus only on extracting cross-domain invariant causal knowledge from the class distribution, effectively preventing excessive transfer of domain-specific information.

C. Causal Learning for Domain Generalization

Causal learning [30] can break the spurious statistical associations between variables that are hard to generalize through causal interventions [31] and further learn the invariant causal relationships between them. Therefore, in recent years, some researchers have explored the potential of causal learning in domain generalization. Among the works related to this article are [32] and [33].

Luo et al. [32] used causal graphs to represent the causal relationships between variables and attempted to implement causal interventions through decoupling learning with VAE. They aimed to decouple semantic factors and variation factors by maximizing the correlation between predictions and semantic factors while minimizing the correlation between predictions and variation factors, thereby enabling the learning of domain-invariant knowledge. However, simply minimizing the correlation between predictions and variation factors can lead to trivial solutions. For example, the model may map the variation factors of different samples to the same random vector, which can minimize the direct association between the variation factors and the predictions. However, at this point, the variation factors no longer contain sample-specific

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information, and the variation factors learned by the model become trivial solutions that are independent of the input, thereby weakening the model's decoupling performance. In contrast, the B-VAE proposed in this article addresses this issue by minimizing the correlation between predictions and domain distribution while maximizing the correlation between domain distribution and inputs.

Chen et al. [33] constructed a virtual domain to simulate domain shifts and utilized meta-learning and causal learning to learn causal meta-knowledge, specifically how to adapt to domain shifts between the source domain and virtual domain during training. This method assumes that noncausal factors can be observed and intervened upon during causal intervention, which often does not hold in real-world scenarios. In contrast, this article proposes a causal intervention mechanism based on the front-door adjustment [34]. By treating class latent variables as the intermediate variable between data and labels, we intervene on this intermediate variable to estimate the true causal effect between data and labels. Maximizing this true causal effect enables the model to capture domain-invariant causal meta-knowledge. In this way, we can precisely capture the causal relationship between data and labels even when noncausal factors cannot be directly observed or intervened upon.

III. Method

A. Framework of the Proposed Method

The training process of CMG is shown in Fig. 1 and mainly consists of four parts.

1) Spatial–Spectral Feature Learning: First, HSI patches are obtained from HSI data as input to the feature extractor. In the feature extractor, the HSI patch is decomposed into spectral sequences and spatial sequences. The spectral encoder captures the dependencies among spectral bands within the HSI patch, and the spatial encoder captures the spatial dependencies among pixels within the HSI patch, resulting in spatial–spectral features.

2) Distribution Learning: First, using the class encoder and the domain encoder of the B-VAE, the spatial–spectral features are decoupled into class distribution and domain distribution. Then, class latent variables are sampled from the class distribution. The forward information bottleneck minimizes the mutual information between class latent variables and spatial–spectral features while maximizing the mutual information between class latent variables and labels, forcing the model to capture critical class discriminative information within the spatial–spectral features. Second, domain latent variables are sampled from the domain distribution, and the reverse information bottleneck maximizes the mutual information between the domain distribution and spatial–spectral features while minimizing the mutual information between the domain distribution and labels, ensuring that the domain distribution accurately captures domain-specific information that is irrelevant to classes. Finally, class latent variables and domain latent variables are sampled from their respective distributions and fed into the decoder for reconstructing the spatial–spectral features, ensuring the distribution modeling capability of the B-VAE.

3) Meta-Task Construction: First, class latent variables are sampled from the class distribution. Then, the domain distribution is perturbed to create a virtual domain distribution, and virtual domain latent variables are sampled from this virtual domain distribution. These class latent variables and virtual domain latent variables are then fed into the decoder of the B-VAE to generate virtual domain features. Through virtual domain generation, domain-generalization meta-tasks are created, consisting of meta-training tasks and meta-validation tasks. The meta-training tasks contain multiple virtual domains to help the model learn domain-invariant causal knowledge, and the meta-validation task contains a virtual domain to verify the model's generalization performance in unseen domains.

4) Dual-Level Optimization: In the inner layer optimization, first, virtual domain features from the meta-validation task are inputted into the class encoder to obtain class latent variables. Second, causal interventions are applied to the class latent variables to derive causal effect vectors. Finally, based on the causal effect vectors, ICEL is computed for a pseudo-update of the model. In the outer optimization, first, class latent variables from the pseudo-updated class encoder are obtained from the meta-validation task's virtual domain features, and a classifier is used for class prediction. Second, generalization risk loss is computed based on class predictions and combined with ICEL for meta-gradient calculation, resulting in meta-gradients. Finally, the model is updated based on the meta gradient.

B. Feature Extractor

Considering that HSI patches have a 3-D structure, and they often have dozens or even hundreds of bands in the spectral dimension, directly modeling their class distribution and domain distribution often comes with high computational costs. Inspired by the latent diffusion model [36], this article uses a feature extractor pretrained on the source domain to map high-dimensional HSI patches to a low-dimensional feature space, obtaining spatial–spectral features. The workflow of the feature extractor is described below.

First, the HSI patch $\boldsymbol{x} \in \mathbb{R}^{H\times W\times B}$ is divided into spectral sequences $\hat{S}=\{\hat{\boldsymbol{s}}_1,\ldots,\hat{\boldsymbol{s}}_B\}$ along the spectral dimension. H, W, and B represent the length, width, and band number of the HSI patch, respectively, $\hat{\boldsymbol{s}}_1 \in \mathbb{R}^{H\times W}$.

Second, the spectral sequence is inputted into a spectral embedding module to capture the dependencies between spectral bands, updating the features of the HSI patch. The updated HSI patch $\hat{\boldsymbol{x}}$ can be represented as

$$\hat{\boldsymbol{x}} = \mathrm{MLP}\big(\mathrm{LN}\big(\mathrm{MSA}\big(\mathrm{LN}\big(\hat{S}\big)\big)+\hat{S}\big)\big)+\mathrm{MSA}\big(\mathrm{LN}\big(\hat{S}\big)\big)+\hat{S}) \quad (1)$$

where $\mathrm{MLP}(\cdot)$ represents a multilayer perceptron, $\mathrm{LN}(\cdot)$ denotes layer normalization, and $\mathrm{MSA}(\cdot)$ represents multihead self-attention.

Third, the HSI patch is divided into spatial sequences along the spatial dimension, and a learnable class token $\widehat{\boldsymbol{s}}$ is added to integrate the global spatial–spectral information of the HSI patch, obtaining spatial sequences $S = \{\widehat{\boldsymbol{s}}; \boldsymbol{s}_1,\ldots,\boldsymbol{s}_{H\times W}\}$, where $\boldsymbol{s}_1 \in \mathbb{R}^B$.

Fourth, spatial sequences are inputted into a spatial embedding module to capture the spatial dependencies between

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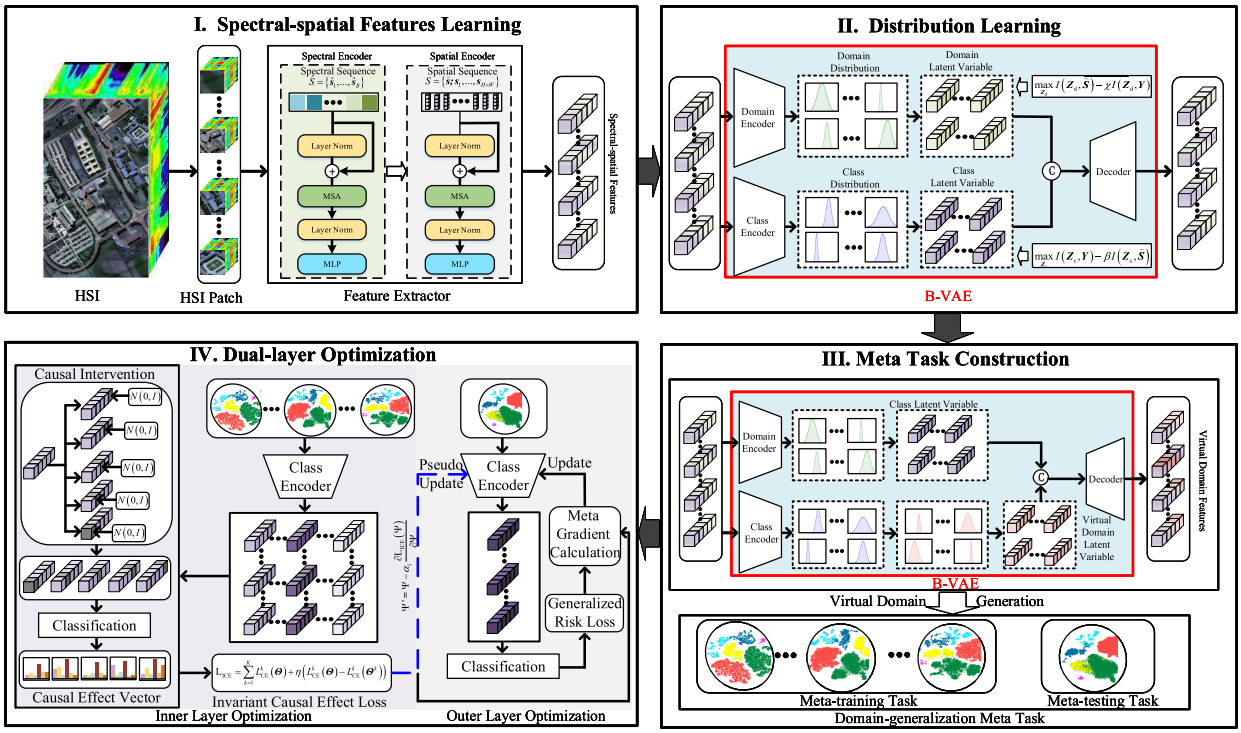


Fig. 1. Framework of CMG.

pixels, updating the spatial sequences. The updated spatial sequences can be represented as

$$\bar{S} = \mathrm{MLP}(\mathrm{LN}(\mathrm{MSA}(\mathrm{LN}(S))+S))+\mathrm{MSA}(\mathrm{LN}(S))+S). \quad (2)$$

The class tokens in the spatial sequences $\bar{S}$ are used as the output, resulting in the spatial–spectral features $\hat{s}$.

C. B-VAE

This section aims to decouple the domain distribution and class distribution in the spatial–spectral features. However, achieving this goal relies on effectively addressing the following issues: 1) How to ensure that the modeling of the class distribution sufficiently captures the class identification information in the spatial–spectral features? 2) How to ensure that the modeling of the domain distribution sufficiently captures the domain-specific information in the spatial–spectral features? 3) How to ensure the sufficient separation of the domain distribution and class distribution?

To address these issues, we propose the B-VAE. First, in the modeling of the class distribution, we construct a forward information bottleneck. By simultaneously minimizing the mutual information between the class distribution and the spatial–spectral features and maximizing its mutual information with the labels, we force the model to focus only on the crucial class identification information in the spatial–spectral features while filtering out domain-specific information irrelevant to the classes. Second, in the modeling of the domain distribution, we construct a reverse information bottleneck. By simultaneously maximizing the mutual information between the domain distribution and the spatial–spectral features and minimizing its mutual information with the labels, we ensure that the domain-specific information in the domain distribution is unrelated to the classes and does not learn trivial solutions unrelated to the input.

This part is organized as follows: First, we present the evidence lower bound (ELBO) of the B-VAE. Second, we derive the upper and lower bounds of the mutual information. Third, we describe the construction process of the forward–reverse information bottleneck. Finally, we present the total loss of this component.

1) ELBO of the B-VAE: In this article, we use the class encoder $q_\Psi(z_c|\hat{s})$ to model the class distribution of the spatial–spectral features and the domain encoder $q_\Phi(z_d|\hat{s})$ to model the domain distribution of the spatial–spectral features. Here, Ψ and Φ are the parameters of the domain encoder and the class encoder, respectively. Second, we sample the class latent variable z_c and domain latent variable z_d from the class distribution and domain distribution, respectively. We concatenate them and input them to the decoder $p_\theta(\hat{s}|z_c, z_d)$ for the reconstruction of the spatial–spectral features, where θ is the parameter of the decoder. Therefore, the following equation holds:

$$p\left(\hat{s}|z_c, z_d\right) = \int\int \sum_C p\left(\hat{s}|z_c, z_d\right) p(z_c)p(z_d)dz_c dz_d. \quad (3)$$

Third, the ELBO of B-VAE can be expressed using Jensens inequality

$$\begin{aligned}\log p\left(\hat{s}\right) &= \log \int\int p\left(\hat{s}, z_c, z_d\right) dz_c dz_d \\ &= \log \int\int p\left(\hat{s} \mid z_c, z_d\right) p(z_c)p(z_d)dz_c dz_d \\ &= \log \mathbb{E}_{q\Psi}\left(z_c|\hat{s}\right), q_\Phi\left(z_d|\hat{s}\right) \frac{p_\theta\left(\hat{s}|z_c, z_d\right)p(z_c)p(z_d)}{q_\Psi\left(z_c, C|\hat{s}\right)q_\Phi(z_d|\hat{s})}\end{aligned}$$

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$$
\begin{aligned}
&\geq \mathbb{E}_{q_\Psi\left(z_c|\widehat{s}\right),q_c\left(z_c|\widehat{s}\right)}\left[\log\frac{p_\theta\left(\widehat{s}|z_c,z_d\right)p\left(z_c\right)p\left(z_d\right)}{q_\Psi\left(z_c|\widehat{s}\right)q_\Phi\left(z_d|\widehat{s}\right)}\right]\\
&= \mathbb{E}_{q_\Psi\left(z_c|\widehat{s}\right),q_\Phi\left(z_d|\widehat{s}\right)}\left[\log p_\theta\left(\widehat{s}|z_c,z_d\right)\right]\\
&\quad+\mathbb{E}_{q_\Psi\left(z_c|\widehat{s}\right),q_\Phi\left(z_d|\widehat{s}\right)}\left[\log\frac{p(z_d)}{q_\Phi\left(z_d|\widehat{s}\right)}\right]\\
&\quad+\mathbb{E}_{q_\Psi\left(z_c|\widehat{s}\right),q_\Phi\left(z_d|\widehat{s}\right)}\left[\log\frac{p(z_c)}{q_{\ \Psi}\left(z_c|\widehat{s}\right)}\right]\\
&= \mathbb{E}_{q_\Psi\left(z_c,C|\widehat{s}\right),q_\Phi\left(z_d|\widehat{s}\right)}\left[\log p_\theta\left(\widehat{s}|z_c,z_d\right)\right]\\
&\quad-D_{\mathrm{KL}}\left(q_\Phi\left(z_d|\widehat{s}\right)\|p(z_d)\right)\\
&\quad-D_{\mathrm{KL}}\left(q_\Psi\left(z_c|\widehat{s}\right)\|p(z_c)\right)
\end{aligned}\tag{4}
$$

where $D_{\mathrm{KL}}(\cdot)$ represents the KL divergence.

In the ELBO mentioned above, $\mathbb{E}_{q_\Psi(z_c|\bar{s}),q_\Phi(z_d|\bar{s})}\left[\log p_\theta(\widehat{s}|z_c,z_d)\right]$ is the reconstruction term. To optimize this term, we utilize the decoder to reconstruct $\widehat{s}$-based z_c and z_d, ensuring the distribution modeling capability of the B-VAE. The goal of the $D_{\mathrm{KL}}(q_\Phi(z_d|\widehat{s})\|p(z_d))$ term is to minimize the KL divergence between the domain posterior distribution and the domain prior distribution $p(z_d)$. In this article, we set the domain prior distribution $p(z_d)$ to $N(0,\boldsymbol{I})$. The goal of the $D_{\mathrm{KL}}(q_\Psi(z_c|\widehat{s})\|p(z_c))$ term is to minimize the KL divergence between the class posterior distribution and the class prior distribution $p(z_c)$. In this article, we obtain a class prior distribution following a mixture of Gaussians by mixing Gaussian distributions $p_{\mathrm{mix}}(z_c,C)$ corresponding to various class prototypes.

Based on the above ELBO, we can obtain the ELBO loss

$$
\begin{aligned}
L_{\mathrm{ELBO}} = &\ \mathrm{MSE}\left(\widehat{s},\bar{s}\right)+D_{\mathrm{KL}}\left(q_\Phi\left(z_d|\widehat{s}\right)\|N(0,\boldsymbol{I})\right)\\
&+D_{\mathrm{KL}}(q_\Psi\left(z_c|\widehat{s}\right)\|p_{\mathrm{mix}}(z_c,C))
\end{aligned}\tag{5}
$$

where $\mathrm{MSE}(\cdot)$ is the mean squared error, $\bar{s}$ is the reconstructed spatial–spectral feature by the decoder.

2) Upper and Lower Bounds of Mutual Information: The forward–reverse information bottleneck requires the calculation of mutual information between variables. However, directly calculating mutual information between variables is challenging. Therefore, in this section, we provide upper and lower bounds of mutual information to lay the foundation for optimizing the forward–reverse information bottleneck information bottleneck.

a) Upper bound of mutual information: For variables $\boldsymbol{A}$ and $\boldsymbol{B}$, the mutual information between them is denoted as

$$
\begin{aligned}
I(\boldsymbol{A},\boldsymbol{B}) &= \int\int p(\boldsymbol{a},\boldsymbol{b})\log\frac{p(\boldsymbol{a},\boldsymbol{b})}{p(\boldsymbol{a})p(\boldsymbol{b})}d\boldsymbol{a}d\boldsymbol{b}\\
&= \int\int p(\boldsymbol{a},\boldsymbol{b})\log\frac{p(\boldsymbol{b}|\boldsymbol{a})}{p(\boldsymbol{b})}d\boldsymbol{a}d\boldsymbol{b}\\
&= \int\int p(\boldsymbol{a},\boldsymbol{b})\log p(\boldsymbol{b}|\boldsymbol{a})d\boldsymbol{a}d\boldsymbol{b}\\
&\quad-\int p(\boldsymbol{b})\log p(\boldsymbol{b})d\boldsymbol{b}.
\end{aligned}\tag{6}
$$

Since $p(\boldsymbol{b})$ is difficult to directly compute, we introduce $r(\boldsymbol{b})$ as the variational approximation of this distribution. Due to $D_{\mathrm{KL}}(p(\boldsymbol{b}),r(\boldsymbol{b}))\geq 0$, we can derive $\int p(\boldsymbol{b})\log p(\boldsymbol{b})d\boldsymbol{b}\geq\int p(\boldsymbol{b})\log r(\boldsymbol{b})d\boldsymbol{b}$. Therefore, the upper bound of $I(\boldsymbol{A},\boldsymbol{B})$ can be obtained as

$$
\begin{aligned}
I(\boldsymbol{A},\boldsymbol{B}) &\leq \int\int p(\boldsymbol{a},\boldsymbol{b})\frac{\log p(\boldsymbol{b}|\boldsymbol{a}))}{r(\boldsymbol{b})}d\boldsymbol{a}d\boldsymbol{b}\\
&= D_{\mathrm{KL}}(p(\boldsymbol{b}|\boldsymbol{a})\|r(\boldsymbol{b})).
\end{aligned}\tag{7}
$$

b) Lower bound of mutual information: For variables $\boldsymbol{A}$ and $\boldsymbol{B}$, the lower bound of the mutual information between them is denoted as

$$
\begin{aligned}
I(\boldsymbol{A},\boldsymbol{B}) &= H(\boldsymbol{B})-H(\boldsymbol{B}|\boldsymbol{A})\\
&= H(\boldsymbol{B})+\mathbb{E}_{\boldsymbol{a}\sim\boldsymbol{A}}\left[\mathbb{E}_{\boldsymbol{b}\sim p(\boldsymbol{b}|\boldsymbol{a})}\left[\log(p(\boldsymbol{b}|\boldsymbol{a}))\right]\right]\\
&= H(\boldsymbol{B})+\mathbb{E}_{\boldsymbol{a}\sim\boldsymbol{A}}\left[\mathbb{E}_{\boldsymbol{b}\sim p(\boldsymbol{b}|\boldsymbol{a})}\left[\log(p(\boldsymbol{b}|\boldsymbol{a}))\right]\right]\\
&\quad-\mathbb{E}_{\boldsymbol{a}\sim\boldsymbol{A}}\left[\mathbb{E}_{\boldsymbol{b}\sim p(\boldsymbol{b}|\boldsymbol{a})}\left[\log(q(\boldsymbol{b}|\boldsymbol{a}))\right]\right]\\
&\quad+\mathbb{E}_{\boldsymbol{a}\sim\boldsymbol{A}}\left[\mathbb{E}_{\boldsymbol{b}\sim p\left(\boldsymbol{b}|\boldsymbol{a}\right)}\left[\log\left(q\left(\boldsymbol{b}|\boldsymbol{a}\right)\right)\right]\right]\\
&= H(\boldsymbol{B})+\mathbb{E}_{\boldsymbol{a}\sim\boldsymbol{A}}\left[D_{\mathrm{KL}}(p(\boldsymbol{b}|\boldsymbol{a})\|q(\boldsymbol{b}|\boldsymbol{a}))\right]\\
&\quad+\mathbb{E}_{\boldsymbol{a}\sim\boldsymbol{A}}\left[\mathbb{E}_{\boldsymbol{b}\sim p(\boldsymbol{b}|\boldsymbol{a})}\left[\log(q(\boldsymbol{b}|\boldsymbol{a}))\right]\right]\\
&\geq H(\boldsymbol{B})+\mathbb{E}_{\boldsymbol{a}\sim\boldsymbol{A}}\left[\mathbb{E}_{\boldsymbol{b}\sim p(\boldsymbol{b}|\boldsymbol{a})}\left[\log(q(\boldsymbol{b}|\boldsymbol{a}))\right]\right].
\end{aligned}\tag{8}
$$

3) Forward–Reverse Information Bottleneck:

a) Forward information bottleneck: Considering that the class distribution should contain as little domain-specific information as possible while containing as much class discriminative information as possible, we have constructed the following forward information bottleneck:

$$
\max_{\boldsymbol{Z}_c} I\left(\boldsymbol{Z}_c,\boldsymbol{Y}\right)-\beta I\left(\boldsymbol{Z}_c,\widehat{\boldsymbol{S}}\right)\tag{9}
$$

where $I\left(\boldsymbol{Z}_c,\boldsymbol{Y}\right)$ represents the mutual information between the class latent variables and the labels, $I\left(\boldsymbol{Z}_c,\widehat{\boldsymbol{S}}\right)$ represents the mutual information between the class latent variables and the spatial–spectral features, and β is a weight factor.

Based on the upper and lower bounds of mutual information, the following lower bound of the forward information bottleneck can be obtained:

$$
\begin{aligned}
\mathrm{FIB} &= I(\boldsymbol{Z}_c,\boldsymbol{Y})-\beta I\left(\boldsymbol{Z}_c,\widehat{\boldsymbol{S}}\right)\\
&\geq H(\boldsymbol{Y})+\mathbb{E}_{z_c\sim\boldsymbol{Z}_c}\left[\mathbb{E}_{\boldsymbol{y}\sim p(\boldsymbol{y}|\boldsymbol{a})}\left[\log\left(q\left(\boldsymbol{y}|z_c\right)\right)\right]\right]\\
&\quad-\beta D_{\mathrm{KL}}(p(z_c|\widehat{s})\|p(z_c)).
\end{aligned}\tag{10}
$$

Since the first term of this lower bound $H(\boldsymbol{Y})$, which is the entropy of the labels, is a constant, it can be ignored during the optimization process. Furthermore, in this article, we use the class encoder to learn the posterior distribution $\log q_\Psi(z_c|\widehat{s})$ to approximate $p(z_c|\widehat{s})$ and use the empirical distribution $p\left(\widehat{s},\boldsymbol{y}\right)=(1/N)\sum_{n=1}^{N}\delta_{\widehat{s}_n}\left(\widehat{s}\right)\delta_{\boldsymbol{y}_n}\left(\boldsymbol{y}\right)$ to approximate $p(\widehat{s},\boldsymbol{y})$. $\widehat{s}_n$ and $\boldsymbol{y}_n$ represent the nth spatial–spectral feature observed by the model and its corresponding label, respectively. Therefore, the final lower bound of the forward information bottleneck can be obtained as

$$
\begin{aligned}
\mathrm{FIB}\geq\frac{1}{N}\sum_{n=1}^{N}\Bigg[&\int q_\Psi\left(z_c|\widehat{s}_n\right)\log q\left(\boldsymbol{y}_n|z_c\right)dz_c\\
&-\beta D_{\mathrm{KL}}(q_\Psi\left(z_c|\widehat{s}_n\right)\|r(z_c))\Bigg].
\end{aligned}\tag{11}
$$

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In summary, based on the lower bound of the forward information bottleneck, the forward information bottleneck loss can be derived as

$$L_{\mathrm{FIB}} = \frac{1}{N}\sum_{n=1}^{N}\mathrm{CrossEntropy}\left(\hat{\boldsymbol{y}}_n, \boldsymbol{y}_n\right) + \beta D_{\mathrm{KL}}\left(q_{\Psi}\left(\boldsymbol{z}_c|\widehat{\boldsymbol{s}}_n\right)\middle\|r\left(\boldsymbol{z}_c\right)\right) \tag{12}$$

where $\hat{\boldsymbol{y}}_n$ represents the class predictions obtained based on the class latent variables, and $\boldsymbol{y}_n$ represents the labels. It should be noted that in this article, we set $r\left(\boldsymbol{z}_c\right)$ to $p_{\min}(z_c, C)$. Therefore, the second term of the forward information bottleneck loss can be consistent with the third term of the ELBO loss, allowing these terms to be merged during the optimization process.

b) Reverse information bottleneck: Considering that the domain distribution should contain as little class discriminative information as possible, on the one hand. On the other hand, it must ensure a strong correlation with the input to prevent it from learning trivial solutions unrelated to the input, we have constructed the following reverse information bottleneck:

$$\max_{\boldsymbol{Z}_d} I\left(\boldsymbol{Z}_d, \widehat{\boldsymbol{S}}\right) - \chi I(\boldsymbol{Z}_d, \boldsymbol{Y}) \tag{13}$$

where $I(\boldsymbol{Z}_d, \widehat{\boldsymbol{S}})$ represents the mutual information between the domain latent variables and the spatial–spectral features, $I(\boldsymbol{Z}_d, \boldsymbol{Y})$ represents the mutual information between the domain latent variables and the labels, and χ is a weight factor.

Based on the upper and lower bounds of mutual information, the following lower bound of the negative information bottleneck can be obtained:

$$\begin{aligned}\mathrm{RIB} &= I(\boldsymbol{Z}_d, \widehat{\boldsymbol{S}}) - \chi I(\boldsymbol{Z}_d, \boldsymbol{Y}) \\ &\geq H\left(\boldsymbol{Z}_d\right) + \mathbb{E}_{\widehat{\boldsymbol{s}}\sim\widehat{\boldsymbol{S}}}\left[\mathbb{E}_{z_d\sim p\left(z_d|\widehat{\boldsymbol{S}}\right)}\left[\log\left(q_{\Gamma}\left(\widehat{\boldsymbol{S}}|z_d\right)\right)\right]\right] \\ &\quad - \chi D_{\mathrm{KL}}(q(\boldsymbol{y}|z_d)\|p(\boldsymbol{y}))\end{aligned} \tag{14}$$

where Γ represents the parameters of the domain decoder. The domain decoder uses the domain latent variables to reconstruct the spatial–spectral features.

Since the first term of this lower bound $H(\boldsymbol{Z}_d)$ can be considered a constant, it can be ignored during the optimization process. Furthermore, in this article, we use the domain encoder to learn the posterior distribution $\log q_{\Phi}(\boldsymbol{z}_d|\widehat{\boldsymbol{s}})$ to approximate $p(\boldsymbol{z}_d|\widehat{\boldsymbol{s}})$ and use the empirical distribution $p(\widehat{\boldsymbol{s}}, \boldsymbol{y}) = (1/N)\sum_{n=1}^{N}\delta_{\widehat{\boldsymbol{s}}_n}(\widehat{\boldsymbol{s}})\delta_{\boldsymbol{y}_n}(\boldsymbol{y})$ to approximate $p(\widehat{\boldsymbol{s}}, \boldsymbol{y})$. Therefore, the final lower bound of the negative information bottleneck can be obtained as

$$\mathrm{RIB} \geq \frac{1}{N}\sum_{n=1}^{N}\left[\int q_{\Phi}\left(\boldsymbol{z}_d|\widehat{\boldsymbol{s}}_n\right)\log q_{\Gamma}\left(\widehat{\boldsymbol{s}}_n|\boldsymbol{z}_d\right)d\boldsymbol{z}_d - \chi D_{\mathrm{KL}}(q(\boldsymbol{y}|\boldsymbol{z}_d)\|r(\boldsymbol{y}))\right]. \tag{15}$$

In summary, based on the lower bound of the negative information bottleneck, the negative information bottleneck loss can be derived as

$$L_{\mathrm{RIB}} = \frac{1}{N}\sum_{n=1}^{N}\mathrm{MSE}\left(\widehat{\boldsymbol{s}}_n^{d}, \widehat{\boldsymbol{s}}_n\right) + \chi D_{\mathrm{KL}}(q_{\Psi}((\boldsymbol{y})|\boldsymbol{z}_d)\|r(\boldsymbol{y})) \tag{16}$$

where $\widehat{\boldsymbol{s}}_n^{d}$ represents the spatial–spectral features reconstructed based on the domain latent variables. In this article, we set $r(\boldsymbol{y})$ to be a uniform distribution $U(0, 1)$.

4) Total Loss of B-VAE: The total loss of B-VAE can be expressed as

$$L_{\text{B-VAE}} = L_{\mathrm{ELBO}} + \lambda_1 L_{\mathrm{FIB}} + \lambda_2 L_{\mathrm{RIB}} \tag{17}$$

where λ_1 and λ_2 are weighting coefficients.

D. Induction of Causal Meta-Knowledge

To induce causal meta-knowledge that can adapt to unseen target domains, we face two core tasks: Minimizing the model's invariant risk across domains and minimizing the model's generalization risk on unseen target domains. The former focuses on the model's learning ability of domain-invariant causal meta-knowledge, while the latter emphasizes the generalization ability of the learned domain-invariant causal meta-knowledge to unseen target domains. These two tasks are intertwined in the optimization process and can produce conflicting optimization gradients.

Therefore, this article develops a framework for the induction of causal meta-knowledge based on minimizing invariant-generalization risk. Specifically, we construct domain generalization meta-tasks that simulate domain generalization scenarios and tailor a dual-layer optimization mechanism (DLOM) for these meta-tasks. In the inner layer's minimization of invariant risk, the model is guided to learn domain-invariant causal meta-knowledge. In the outer layer's minimization of generalization risk, the model needs to strengthen the causal meta-knowledge's generalization ability to unseen target domains, ensuring its applicability in domain generalization tasks.

Next, we will elaborate on meta-task construction and DLOM.

1) Meta-Task Construction: First, we randomly sample a batch of samples from the source domain and use the feature extractor to obtain the corresponding spatial–spectral features. Then, the spatial–spectral features are input into the class encoder of B-VAE to obtain the class distribution $N(\boldsymbol{\mu}_c, \boldsymbol{\sigma}_c^2)$, and class latent variables $\boldsymbol{z}_c \sim N(\boldsymbol{\mu}_c, \boldsymbol{\sigma}_c^2)$ are sampled from the class distribution. $\boldsymbol{\mu}_c$ and $\boldsymbol{\sigma}_c$ are the mean and variance of the class distribution, respectively. Second, the spatial–spectral features are input into the domain encoder of B-VAE to obtain the domain distribution $N(\boldsymbol{\mu}_d, \boldsymbol{\sigma}_d^2)$, and the domain distribution is perturbed to obtain the virtual domain distribution $N(\boldsymbol{\mu}_v, \boldsymbol{\sigma}_v^2)$. $\boldsymbol{\mu}_v$ and $\boldsymbol{\sigma}_v^2$ are the mean and variance of the virtual domain distribution, respectively, and can be represented as

$$\boldsymbol{\mu}_v = \boldsymbol{\mu}_d + \boldsymbol{\mu}_{\varepsilon}, \boldsymbol{\sigma}_v^2 = \boldsymbol{\sigma}_d^2 + \boldsymbol{\sigma}_{\varepsilon}^2 \tag{18}$$

where $\boldsymbol{\mu}_{\varepsilon}$ and $\boldsymbol{\sigma}_{\varepsilon}^2$ are random variables sampled from Gaussian distributions.

Third, sample virtual domain latent variables $\boldsymbol{z}_d^v \sim N(\boldsymbol{\mu}_v, \boldsymbol{\sigma}_v^2)$ from the virtual domain distribution. Then, concatenate the class latent variables and the virtual domain latent variables, and input them into the decoder $\mathrm{Decoder}_{\theta}(\cdot)$ of the B-VAE to generate the virtual domain features $\widehat{\boldsymbol{s}}_v$:

$$\widehat{\boldsymbol{s}}_v = \mathrm{Decoder}_{\theta}\left(\mathrm{CONCAT}\left(\boldsymbol{z}_c, \boldsymbol{z}_d^v\right)\right) \tag{19}$$

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where CONCAT$(\cdot)$ denotes the concatenation operation.

Finally, randomly select groups of virtual domain features as virtual source domains to construct meta-training tasks, simulating potential domain distribution changes in the real world. Choose the remaining group of virtual domain features as a virtual target domain to construct meta-validation tasks, simulating unseen target domains in the real world.

2) Dual-Layer Optimization Mechanism:

a) Inner layer optimization: Although B-VAE decouples the domain distribution and the class distribution, it can only alleviate the differences in marginal distributions between different domains and is difficult to handle the differences in conditional distributions between them. To address this issue, we designed an inner layer optimization mechanism based on causal learning, which cuts off the false statistical associations between data and labels through causal intervention, thereby achieving accurate induction of cross-domain-invariant causal knowledge.

Specifically, first, consider that the differences in conditional distributions of the same class in different domains are often induced by noncausal factors. For example, in the task of identifying "cats," "cats" in different domains can exhibit significantly different conditional distributions due to color differences. At this point, existing deep learning models based on statistical association modeling often mistakenly establish a statistical association between noncausal factors such as "color" and "cats," and this kind of statistical association is often difficult to generalize to other domains, which limits their generalization performance in cross-domain scenarios. For example, a model trained on white "cats" may perform poorly on black "cats."

If the false statistical association between noncausal factors and labels can be cut off, the impact of differences in conditional distributions between different domains on the model's generalization performance can be reduced. By learning domain-invariant causal knowledge between data and labels, the model's cross-domain generalization ability can be improved. However, noncausal factors are often mixed with causal factors, making them difficult to observe directly. Therefore, we introduce a structural causal model (SCM) as shown in Fig. 2 and use the front-door criterion to adjust for causal intervention [37], to calculate the true causal effect between the spectral feature and the label

$$p\left(\boldsymbol{Y}\,|\,\mathrm{do}\left(\widehat{\boldsymbol{S}}\right)\right)=\sum_{\widehat{s}}\sum_{z_c}p(\boldsymbol{Y}\,|\,\mathrm{do}(\boldsymbol{Z}_c=\boldsymbol{z}_c))\times p\left(\boldsymbol{Z}_c=\boldsymbol{z}_c\,|\,\widehat{\boldsymbol{s}}\right)p(\widehat{\boldsymbol{s}}) \tag{20}$$

where do$(\cdot)$ denotes the causal intervention operation. In this work, we input the class encoder with each virtual domain feature from the meta-training task to obtain the class latent variable $\boldsymbol{z}_c$. Then, we intervene on $\boldsymbol{z}_c$ by replacing each dimension of $\boldsymbol{z}_c$ with a randomly sampled vector from a Gaussian distribution.

Second, we integrate the class predictions obtained after intervening on each dimension of $\boldsymbol{z}_c$ for the causal effect vector to obtain the causal effect vector $\boldsymbol{\pi}$

$$\boldsymbol{\pi}=\mathrm{CONCAT}\left(\varsigma\left(\hat{\boldsymbol{z}}_{c,l}\right),\ldots,\varsigma\left(\hat{\boldsymbol{z}}_{c,L}\right)\right) \tag{21}$$

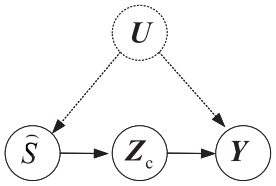


Fig. 2. SCM. $\widehat{s}$ represents feature, z_c is the class latent variable sampled from the class distribution, $\boldsymbol{Y}$ is the label, and $\boldsymbol{U}$ is the unobserved noncausal factor.

where $\hat{\boldsymbol{z}}_{c,l}$ represents the class latent variable with the lth dimension replaced, L is the number of dimensions in $\boldsymbol{z}_c$, and $\varsigma(\cdot)$ is the classifier.

Third, we treat the causal effect vectors corresponding to the same-class virtual source-domain features in the meta-training task as positive sample pairs and those corresponding to different-class virtual domain features as negative sample pairs. We use the causal effect loss to minimize the difference between positive sample pairs and simultaneously maximize the difference between positive and negative sample pairs, enhancing the intraclass consistency and interclass distinctiveness of the class latent variables corresponding to each virtual domain feature pair. The causal effect loss is

$$L_{\mathrm{CE}}=-\frac{1}{Q}\sum_{q=1}^{Q}\left[\log\frac{\exp\left(\mathrm{sim}\left(\boldsymbol{\pi}_q^t,\boldsymbol{\pi}_{q+}^t\right)/\tau\right)}{\exp\left(\left(\mathrm{sim}\left(\boldsymbol{\pi}_q^t,\boldsymbol{\pi}_{q+}^t\right)/\tau\right)+\sum_{j=1}^{J}\exp\left(\mathrm{sim}\left(\boldsymbol{\pi}_q^t,\boldsymbol{\pi}_{q-j}^t\right)/\tau\right)\right)}\right] \tag{22}$$

where Q denotes the number of virtual source-domain features in the meta-training task, $\boldsymbol{\pi}_q^t$ is the causal effect vector corresponding to the qth virtual source-domain feature in the meta-training set, and $\boldsymbol{\pi}_{q+}^t$ and $\boldsymbol{\pi}_{q\text{-}j}^t$, respectively, represent the positive and negative samples of $\boldsymbol{\pi}_q$. J is the number of negative samples, sim$(\cdot)$ represents the similarity measure, here being the Euclidean distance, and τ is the temperature coefficient.

Fourth, we combine the causal effect loss and invariant risk minimization [38] to design the ICEL

$$L_{\mathrm{ICE}}=\sum_{k=1}^{K}L_{\mathrm{CE}}^k(\boldsymbol{\Theta})+\eta\left(L_{\mathrm{CE}}^k(\boldsymbol{\Theta})-L_{\mathrm{CE}}^k\left(\boldsymbol{\Theta}^k\right)\right) \tag{23}$$

where $L_{\mathrm{CE}}^k(\boldsymbol{\Theta})$ represents the causal effect loss of the model on the kth virtual source domain when the positive and negative sample selection ranges over all virtual source domain features, and $L_{\mathrm{CE}}^k(\boldsymbol{\Theta}^k)$ represents the causal effect loss of the model on the kth virtual source domain when the positive and negative sample selection ranges over the kth virtual source domain. In this way, the model is encouraged to learn domain-invariant causal knowledge by minimizing its invariant risk across virtual domains during the inner layer optimization process.

Finally, pseudo-update the class encoder based on the ICEL

$$\boldsymbol{\Psi}'=\boldsymbol{\Psi}-\alpha_1\nabla\frac{\partial L_{\mathrm{ICE}}(\boldsymbol{\Psi})}{\partial\boldsymbol{\Psi}} \tag{24}$$

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TABLE I
CLASSES AND THEIR NUMBER OF THE PAVIA DATASET

Pavia University (PaviaU)			Pavia Center (PaviaC)		
No	Class	Number	No	Class	Number
P1	Tree	3064	P1	Tree	7598
P2	Asphalt	6631	P2	Asphalt	9248
P3	Brick	3682	P3	Brick	2685
P4	Bitumen	1330	P4	Bitumen	7287
P5	Shadow	947	P5	Shadow	2863
P6	Meadow	18649	P6	Meadow	3090
P7	Bare soil	5029	P7	Bare soil	6584
	Total	39332		Total	39355

where $\boldsymbol{\Psi}'$ is the class encoder parameters after pseudo-updating, $\boldsymbol{\Psi}$ is the original parameters of the class encoder, α_1 is the inner learning rate, and $\nabla(\partial L_{\mathrm{ICE}}(\boldsymbol{\Psi})/\partial\boldsymbol{\Psi})$ is the inner layer optimization gradient.

b) Outer-layer Optimization: The outer-layer optimization aims to evaluate the model's generalization performance on unseen target domains. Specifically, first, we calculate the generalization risk loss L_{GR} of the pseudo-updated model on the meta-validation task's virtual target domain.

$$L_{\mathrm{GR}} = \mathrm{CrossEntropy}\left(\tilde{\boldsymbol{y}}_v, \breve{\boldsymbol{y}}_v\right) \tag{25}$$

where $\tilde{\boldsymbol{y}}_v$ and $\breve{\boldsymbol{y}}_v$ are the predicted and true labels of the virtual target-domain features, respectively. In the meta-validation task, predicted labels can be obtained by inputting the class latent variables corresponding to the virtual target-domain features into a classifier.

Second, the meta-gradient $\nabla\boldsymbol{G}_M$ is calculated based on the invariant risk and generalization risk losses

$$\nabla\boldsymbol{G}_M = \frac{\partial L_{\mathrm{GR}}(\boldsymbol{\Psi}') + \partial L_{\mathrm{ICE}}(\boldsymbol{\Psi})}{\partial\boldsymbol{\Psi}}. \tag{26}$$

Finally, the original parameters of the class model are updated based on the meta-gradient

$$\hat{\boldsymbol{\Psi}} = \boldsymbol{\Psi} - \alpha_2\nabla\boldsymbol{G}_M \tag{27}$$

where $\hat{\boldsymbol{\Psi}}$ is the meta-updated model parameters and α_2 is the meta-learning rate. This approach allows the model to balance between inductive ability to domain-invariant knowledge and generalization ability to unseen target domains. It enables the model to induce causal meta-knowledge shared across tasks from a large number of meta-tasks, thereby improving its performance in domain generalization tasks.

IV. EXPERIMENT

A. Dataset

1) Pavia Dataset: This dataset consists of two subdatasets: The Pavia University (PaviaU) dataset and the Pavia Center (PaviaC) dataset. The PaviaU dataset contains pixels and 103 spectral bands, while the PaviaC dataset contains pixels and 102 spectral bands. To ensure that both datasets have the same number of spectral bands, we followed the experimental settings of [35] and removed the last spectral band from the Pavia dataset. The classes and their number in the dataset are shown in Table I.

TABLE II
CLASSES AND THEIR NUMBER OF THE HOUSTON DATASET

Houston 2013 (HS13)			Houston 2018 (HS18)		
No	Class	Number	No	Class	Number
H1	Grass healthy	345	H1	Grass healthy	1353
H2	Grass stressed	365	H2	Grass stressed	4888
H3	Trees	365	H3	Trees	2766
H4	Water	285	H4	Water	22
H5	Residential buildings	319	H5	Residential buildings	5347
H6	Non-residential buildings	480	H6	Non-residential buildings	32459
H7	Road	443	H7	Road	6365
	Total	2530		Total	53200

TABLE III
CLASSES AND THEIR NUMBER OF THE GID DATASET

GIDWH			GIDDZ		
No	Class	Number	No	Class	Number
G1	Rural residential	14418	G1	Rural residential	7608
G2	Graden land	683	G2	Graden land	10626
G3	Irrigated field	3819	G3	Irrigated field	12735
G4	Road	17776	G4	Road	22824
G5	Lake	18774	G5	Lake	472
	Total	55470		Total	54220

2) Houston Dataset: We utilized the high-resolution hyperspectral datasets collected over the University of Houston campus and its surrounding areas: Houston 2013 (HS13) and Houston 2018 (HS18). The HS13 dataset consists of 144 spectral bands covering the wavelength range from 380 to 1050 nm, with a spatial resolution of 2.5 m. The HS18 dataset consists of 48 spectral bands covering the same wavelength range, with a spatial resolution of 1 m. Following the experimental settings of [35], we selected pixels from each dataset corresponding to the same scene, which includes seven common classes. During training, only the 48 bands from the HS13 dataset were used to ensure consistency between the source and target domains. The classes and their number in the datasets are shown in Table II.

3) GID Dataset: This dataset [43] is a large-scale remote sensing dataset constructed by Wuhan University, consisting of images captured at different times in various regions of China. In this article, we selected GIDWH, collected in Wuhan, Hubei, and GIDDZ, collected in Danzhou, Hainan, for our experiments. GIDWH consists of 345 × 2190 pixels covering five classes. GIDDZ consists of 690 × 730 pixels with the same classes as GIDWH. The specific class types and quantities for GIDWH and GIDDZ are shown in Table III.

B. Experimental Setting

All experiments were conducted on a computer equipped with an Intel Core i7-10700KF CPU running at 3.80 GHz, 32 GB of RAM, and an RTX 3060Ti GPU. The experiments were implemented in PyTorch. The comparative methods in this study include noncross-domain methods: graph generative structure-aware Transformer (GraphGST) [39], heterogeneous transfer learning based on the CNN (HTCNN) [40], SpectralFormer [41], HSI Transformer (HiT) [42]; domain generalization methods of HSIs: language-aware domain generalization network (LDGnet) [35], SDEnet [14]; domain generalization methods in the field of computer vision: Stylization and destylization (StyDeSty) [44], meta-causal

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TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (PAVIA DATASET)

Class	HTCNN	SpectralFormer	HiT	GraphGST	SDEnet
P1	90.76±3.25	82.71±3.54	85.96±2.51	92.60±1.32	84.93±3.15
P2	96.73±2.12	86.69±1.85	88.73±1.52	96.71±0.15	95.38±1.03
P3	74.71±4.35	**79.29±2.15**	74.26±4.53	52.89±6.85	73.52±3.46
P4	20.34±8.12	49.54±6.12	41.06±5.78	53.11±4.26	72.54±1.07
P5	99.97±0.01	98.74±1.02	99.65±0.21	99.69±0.12	99.72±0.15
P6	75.99±2.12	**92.43±1.23**	90.94±1.35	79.74±3.23	76.41±3.86
P7	72.60±3.12	64.85±4.36	65.64±4.21	76.63±4.52	67.44±6.85
OA (%)	74.50±2.42	76.21±2.63	75.49±2.01	79.20±1.85	81.79±1.53
AA (%)	75.87±3.30	79.19±2.89	78.04±2.87	78.05±2.92	81.42±2.80
Kappa (%)	69.13±1.93	71.79±2.12	70.66±1.75	74.86±1.62	78.15±1.96
Class	LDGnet	StyDeSty	MCL	AdvST	CMG
P1	91.24±1.21	94.41±1.34	94.85±1.65	86.56±1.25	**95.07±1.07**
P2	97.56±0.54	97.34±1.03	95.42±1.08	97.46±0.51	**97.57±0.89**
P3	62.76±3.24	44.92±5.86	53.33±3.65	69.98±2.12	54.12±4.36
P4	57.94±4.85	54.11±7.88	49.80±3.95	59.65±5.26	**72.64±3.12**
P5	99.72±0.21	**100±0**	99.44±0.15	99.97±0.01	**100±0**
P6	77.77±4.55	77.51±3.12	75.63±2.12	84.43±2.74	78.80±2.15
P7	81.08±3.12	**89.48±1.02**	84.89±1.35	69.29±3.53	80.59±3.01
OA (%)	82.47±2.45	82.51±2.14	80.97±1.51	80.93±2.03	**85.37±1.42**
AA (%)	81.15±2.53	79.68±2.89	79.05±1.98	81.05±2.20	**82.68±2.08**
Kappa (%)	78.80±1.87	78.77±1.75	76.90±1.23	77.04±1.72	**82.29±1.09**

TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (HS13–HS18 DATASETS)

Class	HTCNN	SpectralFormer	HiT	GraphGST	SDEnet
H1	62.01±6.42	54.25±5.14	44.94±6.54	67.70±4.12	74.87±2.18
H2	74.33±3.54	64.51±3.40	73.04±3.23	81.57±3.64	78.91±1.36
H3	42.12±4.52	50.04±5.87	55.53±1.25	53.04±2.71	55.42±8.51
H4	86.36±3.21	81.82±3.17	**100±0**	90.91±1.31	95.45±2.12
H5	73.82±2.85	72.85±2.51	**74.14±7.26**	69.55±2.33	72.77±3.23
H6	87.01±3.47	88.77±1.84	88.91±1.35	86.01±1.74	88.40±2.59
H7	44.89±8.12	53.95±6.19	52.65±5.81	**61.82±7.12**	53.31±4.85
OA (%)	76.51±3.42	77.88±3.97	78.78±3.37	78.87±2.93	79.70±2.15
AA (%)	67.22±4.29	66.60±3.97	69.89±3.63	72.94±3.28	74.16±3.55
Kappa (%)	59.58±2.86	61.58±2.68	63.34±3.01	64.64±2.55	65.30±1.91
Class	LDGnet	StyDeSty	MCL	AdvST	CMG
H1	70.36±6.32	75.24±4.85	73.69±3.86	56.76±8.74	**78.57±2.84**
H2	77.93±2.51	81.42±3.25	77.19±5.12	77.86±4.12	**83.84±3.01**
H3	57.45±3.42	55.21±2.12	54.70±3.02	57.99±5.11	**60.30±2.12**
H4	**100±0**	95.46±1.34	81.82±4.36	86.36±4.32	95.45±3.21
H5	70.92±4.36	75.46±3.12	72.00±4.21	73.29±2.34	69.98±4.23
H6	89.82±1.24	87.03±3.55	89.36±3.44	88.60±4.16	**90.25±2.41**
H7	52.54±4.26	53.31±4.77	50.53±2.34	55.49±5.41	55.40±2.11
OA (%)	80.19±2.31	79.37±2.35	79.65±2.43	79.71±2.71	**81.60±1.85**
AA (%)	74.14±3.16	74.73±3.29	71.33±3.76	70.91±4.89	**76.26±2.85**
Kappa (%)	65.77±1.89	65.16±1.96	64.83±2.16	65.31±2.31	**68.37±1.76**

learning (MCL) [33], and adversarial learning with semantics transformations (AdvST) [45].

To ensure a fair comparison, in all experiments, we assumed that the target domain was unseen and only trained the model using data from the source domain, testing it using data from the target domain. During the meta-learning process, when training CMG, we used AdamW as the optimizer, set the number of episodes to 2000, the inner learning rate to 0.001, the meta-learning rate to 0.001, the patch size to 11, and the number of inner layer optimization steps to 5. The feature extractor included five stacked sets of spectral embedding modules and spatial embedding modules. To eliminate the influence of randomness on the experiments, all results are the average of ten runs. Three evaluation metrics were chosen, including Overall Accuracy (OA), Average Accuracy (AA) per class, and Kappa coefficient.

C. Comparative Experiments

The experimental results of CMG and its comparative methods on two datasets are shown in Tables IV and V. From these tables, we can observe the following points.

1) Compared to Noncross-Domain Methods: CMG achieves higher classification performance. This is because noncross-domain methods adopt a learning paradigm based on empirical risk minimization, facing the challenge of cross-domain knowledge transfer in cross-domain scenarios, that is, performing well in the source domain but limited in generalization performance in the target domain. In contrast, CMG captures domain-invariant causal knowledge precisely by minimizing the model's invariant risk in each virtual source domain, ensuring its robust generalization ability in cross-domain scenarios.

2) Compared to SDEnet: CMG achieves classification performance improvements of 3.58% and 1.90% on the two datasets, respectively. This is because SDEnet considers morphological structural information as domain-invariant representation, which may be overly idealistic. Due to differences in the collection areas of different HSIs, the spatial structures of the same land-cover types may exhibit significant differences. In contrast, CMG does not rely on ideal assumptions about the spatial structures of HSI land-cover types but rather induces domain-invariant causal meta-knowledge in the domain generalization scenario task, enabling it to adapt to changes in the morphological structures of the same land-cover types across different domains.

3) Compared to LDGnet: CMG achieves classification performance improvements of 2.90% and 1.41% on the two datasets, respectively. This is because, although LDGnet alleviates the conditional distribution differences between the source and target domains by introducing linguistic modality knowledge, the process of aligning linguistic and visual modality features leads the model to slightly sacrifice class discriminability to enhance feature sharing. In contrast, CMG benefits from its DLOM based on invariant-generalization risk minimization, balancing both the domain-invariant nature and class discriminability of features.

4) Compared to MCL: CMG demonstrates stronger domain generalization performance. This is because MCL assumes that noncausal factors can be observed and intervened upon during causal intervention. In MCL, the model's ability to learn causal meta-knowledge depends on human-defined intervention rules. However, fixed predefined rules often struggle to fully capture the complex causal relationships between data and labels. In contrast, CMG introduces a causal intervention mechanism based on the front-door adjustment [34], where class latent variables are treated as intermediate variables between data and labels. By intervening on these intermediate variables, the causal effect between data and labels is estimated. Since these intermediate variables are learnable, intervening in them helps the model better capture the complex causal relationships between data and labels.

5) Compared to AdvST and StyDeSty: CMG demonstrates stronger domain generalization performance. This is because both AdvST and StyDeSty attempt to expand the source-domain distribution through data augmentation. However, they face a tradeoff between diversity and authenticity: During distribution expansion, they may harm the authenticity

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TABLE VI

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE GIDDZ-GIDWH DATASET

Dataset	Index	HTCNN	SpectralFormer	HiT	GraphGST	SDEnet
GIDDZ-GIDWH	OA (%)	63.79±2.65	66.81±2.23	67.12±2.54	68.76±2.13	71.71±1.69
	AA (%)	60.59±3.95	67.93±4.81	66.55±4.03	64.66±4.32	67.48±3.66
	Kappa (%)	50.23±2.36	53.94±2.03	53.40±2.28	55.38±1.87	59.24±1.42
	Index	LDGnet	StyDeSty	MCL	AdvST	CMG
	OA (%)	71.23±1.34	70.55±1.72	72.06±1.45	71.88±1.58	**73.95±1.23**
	AA (%)	69.70±3.48	70.82±3.75	73.47±3.81	71.78±3.54	**75.79±3.12**
	Kappa (%)	58.59±1.19	58.00±1.28	60.35±1.21	59.74±1.32	**64.10±1.04**

TABLE VII

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE GIDWH-GIDDZ DATASET

Dataset	Index	HTCNN	SpectralFormer	HiT	GraphGST	SDEnet
GIDWH-GIDDZ	OA (%)	71.86±2.65	73.33±1.98	74.90±2.87	74.30±2.34	76.34±2.12
	AA (%)	65.14±3.96	65.11±4.87	63.12±4.67	67.53±4.35	64.23±4.02
	Kappa (%)	60.39±2.33	63.60±1.65	65.27±2.43	65.29±1.95	67.90±1.87
	Index	LDGnet	StyDeSty	MCL	AdvST	CMG
	OA (%)	75.10±1.96	76.46±1.46	76.23±1.24	75.29±1.58	**78.85±1.38**
	AA (%)	66.42±3.74	65.10±3.97	65.75±4.66	63.03±4.42	**72.06±3.24**
	Kappa (%)	66.02±1.75	67.61±1.17	67.39±1.03	66.11±1.21	**70.57±1.15**

TABLE VIII

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE HS13–HS18 DATASET

Dataset	Index	HTCNN	SpectralFormer	HiT	GraphGST	SDEnet
HS13-HS18	OA (%)	76.51±3.42	77.88±3.97	78.78±3.37	78.87±2.93	79.70±2.15
	AA (%)	67.22±4.29	66.60±3.97	69.89±3.63	72.94±3.28	74.16±3.55
	Kappa (%)	59.58±2.86	61.58±2.68	63.34±3.01	64.64±2.55	65.30±1.91
	Index	LDGnet	StyDeSty	MCL	AdvST	CMG
	OA (%)	80.19±2.31	79.37±2.35	79.65±2.43	79.71±2.71	**81.60±1.85**
	AA (%)	74.14±3.16	74.73±3.29	71.33±3.76	70.91±4.89	**76.26±2.85**
	Kappa (%)	65.77±1.89	65.16±1.96	64.83±2.16	65.31±2.31	**68.37±1.76**

TABLE IX

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE HS18–HS13 DATASET

Dataset	Index	HTCNN	SpectralFormer	HiT	GraphGST	SDEnet
HS18-HS13	OA (%)	85.46±3.71	87.00±3.52	88.10±3.45	88.42±3.18	90.20±2.42
	AA (%)	86.12±4.59	87.12±3.86	87.55±4.12	86.57±3.87	89.95±4.53
	Kappa (%)	82.97±3.22	84.78±2.94	86.06±3.19	86.41±2.79	88.52±1.95
	Index	LDGnet	StyDeSty	MCL	AdvST	CMG
	OA (%)	91.23±1.66	89.49±2.12	90.51±1.71	90.40±1.85	**92.69±1.32**
	AA (%)	91.28±3.06	88.69±4.01	90.13±3.42	89.81±3.84	**92.47±2.74**
	Kappa (%)	89.72±1.31	87.67±1.73	88.89±1.36	88.74±1.44	**91.44±1.03**

of the augmented data by overly pursuing diversity, or compromise diversity by focusing too much on authenticity. In contrast, CMG decouples domain distribution from class distribution, perturbing only the domain distribution during virtual domain generation. This method allows CMG to expand diversity while maintaining the authenticity of virtual domain features.

In summary, compared to the state-of-the-art noncross-domain methods and domain generalization methods, CMG demonstrates the strongest domain generalization performance, validating its superiority in HSI domain generalization tasks.

D. Adaptability Experiment of Spatial and Temporal Shifts

1) Spatial Shift: We have added experiments on the GIDWH-GIDDZ and GIDDZ-GIDWH datasets to validate the domain generalization performance of CMR and its comparison methods in handling severe spatial shifts. GIDWH and GIDDZ were collected in Wuhan, Hubei, and Danzhou, Hainan, respectively, making them well-suited for evaluating the model's generalization performance across nonoverlapping geographical regions.

The experimental results of CMR and its comparison methods on the GIDWH-GIDDZ and GIDDZ-GIDWH datasets are shown in Tables VI and VII. From the results, it can be observed that the geographical locations of the source and target domains being nonoverlapping, the differences in their geographical environments, land-cover distributions, and imaging conditions lead to significant cross-domain distribution shifts. Nevertheless, CMR consistently demonstrated the best classification performance among all comparison methods, validating its effectiveness in domain generalization tasks.

2) Temporal Shift: We have added experiments on the HS18–HS13 (HS18–HS13) dataset to evaluate the domain generalization performance of the proposed method in scenarios with temporal shifts. The experimental results of CMR and its comparison methods on the HS13–HS18 and HS18–HS13 datasets are shown in Tables VIII and IX. The results indicate that the temporal shift between the source and target domains, the dynamic environmental changes, variations in lighting conditions, and the evolution of land cover increase the difficulty of domain generalization. Nevertheless, CMR consistently outperformed all comparison methods, further validating its effectiveness in domain generalization tasks.

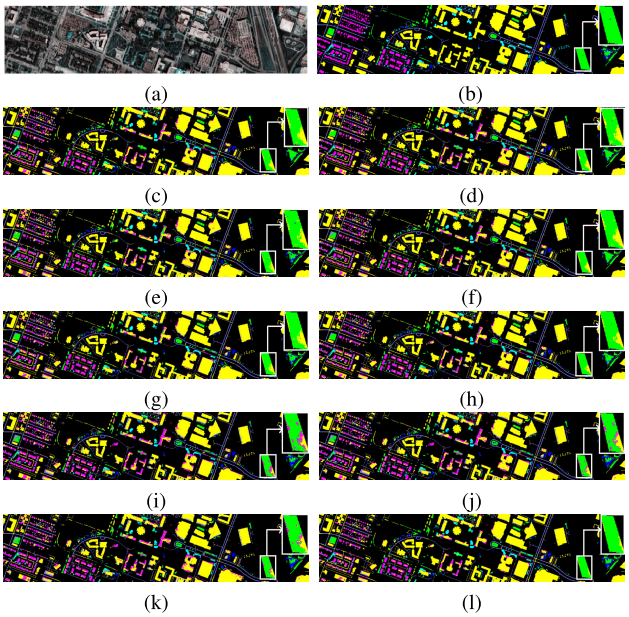


Fig. 3. Classification accuracy map on HS18 dataset. (a) False-color image. (b) Ground truth. (c) HTCNN. (d) SpectralFormer. (e) HiT. (f) GraphGST. (g) SDEnet. (h) LDGnet. (i) StyDeSty. (j) MCL. (k) AdvST. (l) CMR.

E. Visual Analysis

1) Classification Accuracy Maps: In this section, we visualize the classification accuracy maps of CMG and its comparative methods on the Pavia dataset and the Houston dataset, as shown in Figs. 3 and 4. We have labeled the GT and method names above or below the corresponding visualization results and highlighted certain areas with boxes (white lines), enlarging them to demonstrate the superiority of the proposed method. We can see that CMG exhibits the best continuity in classification accuracy on both datasets.

2) t-SNE Maps: In this section, we present t-SNE maps of the original HSI and the class latent variables output by CMG

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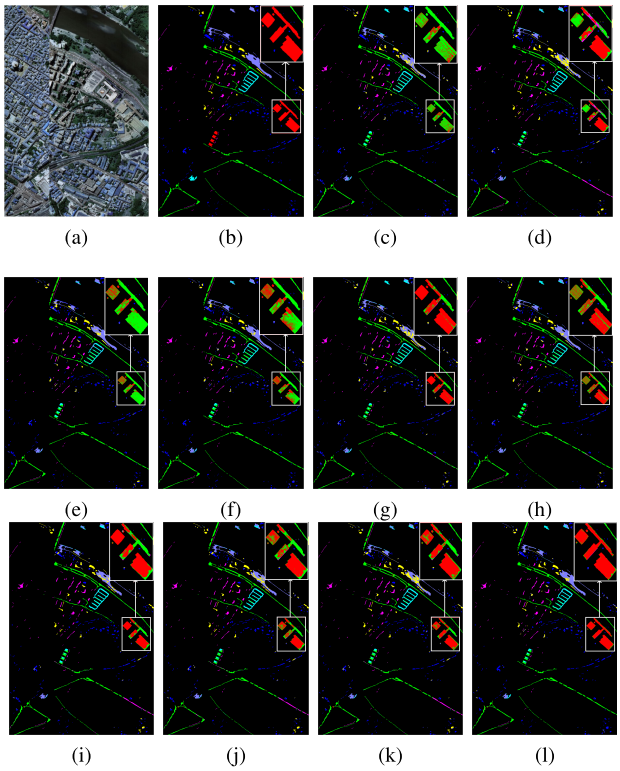


Fig. 4. Classification accuracy map on PaviaC dataset. (a) False-color image. (b) Ground truth. (c) HTCNN. (d) SpectralFormer. (e) HiT. (f) GraphGST. (g) SDEnet. (h) LDGnet. (i) StyDeSty. (j) MCL. (k) AdvST. (l) CMR.

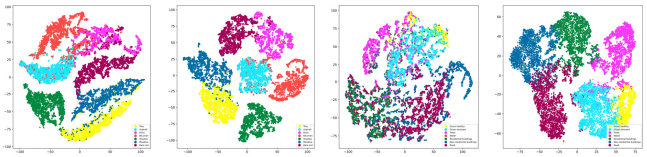


Fig. 5. t-SNE map. (a) t-SNE map of the original HSI on the PaviaC dataset. (b) t-SNE map of the class latent variable learned by CMG on the PaviaC dataset. (c) t-SNE map of the original HSI on the HS18 dataset. (d) t-SNE map of the class latent variable learned by CMG on the HS18 dataset.

in Fig. 5. From these figures, we can observe that the t-SNE map of the original HSI shows low intraclass consistency and interclass separability. In contrast, CMG, through the induction of causal meta-knowledge, accurately captures the key class identification information in the HSI. Its output class latent variables exhibit higher intraclass consistency and interclass separability.

F. Ablation Study

CMG consists of three key components: B-VAE, ICEL, and DLOM. We constructed ablation models as shown in Table X and conducted ablation experiments to verify the effectiveness of each component, as shown in Table XI. Below, we will introduce the construction principles of each ablation model.

1) *CMG-A:* This model uses a feature extractor to obtain the spatial–spectral features of the source-domain HSI and updates the model using cross-entropy loss. After training, it directly classifies the target-domain HSI.
2) *CMG-B:* Building upon CMG-A, this model adds the B-VAE. After training, it uses the class encoder to obtain the class distribution of the target domain's spatial–spectral features and samples the class latent variables from the distribution. It then completes the target-domain classification task based on the class latent variables.
3) *CMG-C:* Building upon CMG-B, this model adds the ICEL. It uses the B-VAE to generate virtual domain features and optimizes the model using the ICEL.
4) *CMG:* Building upon CMG-C, this model adds the DLOM.

TABLE X
ABLATION MODELS

Component	CMG-A	CMG-B	CMG-C	CMG
B-VAE	✘	✔	✔	✔
ICEL	✘	✘	✔	✔
DLOM	✘	✘	✘	✔

TABLE XI
IMPACT OF DIFFERENT COMPONENTS ON OA (%)

Dataset	CMG-A	CMG-B	CMG-C	CMG
PaviaU-PaviaC	78.56±2.04	81.12±1.75	83.73±1.51	**85.37±1.42**
HS18-HS13	87.45±1.85	89.15±1.54	91.48±1.44	**92.69±1.32**
HS13-HS18	76.86±2.11	78.53±2.21	80.86±2.03	**81.60±1.85**
GIDDZ-GIDWH	66.93±2.85	69.45±1.58	71.31±2.31	**73.95±1.23**
GIDWH-GIDDZ	73.12±2.55	75.35±2.44	77.48±1.56	**78.85±1.38**

The classification performance of each ablation model on the Pavia dataset is shown in Table VII. From the results, we can see the following.

1) Compared to CMG-A, CMG-B achieved an improvement in classification accuracy on the five datasets. This is because CMG-A updates the model by minimizing the cross-entropy loss of the source domain, which makes it difficult to handle the data distribution differences between the source and target domains. The B-VAE, on the other hand, decouples the class distribution and domain distribution by using the forward–reverse information bottleneck, promoting cross-domain knowledge transfer. In particular, the forward information bottleneck compresses information flow, forcing the model to focus only on the key class identification information, filtering out noise information and domain-specific information unrelated to the class, and ensuring the model's adaptability to domain distribution changes.

2) Compared to CMG-B, CMG-C achieved an improvement in classification accuracy on the five datasets. This is because CMG-B can only reduce the marginal distribution difference between the source and target domains, but it is difficult to handle the conditional distribution difference between them. In contrast, CMG-C, by minimizing the model's invariant risk on virtual source domains, can exploit domain-invariant causal knowledge shared across different domains, improving the model's adaptability to cross-domain scenarios.

3) Compared to CMG-C, CMG achieved an improvement in classification accuracy on the five datasets. This is because CMG uses a DLOM, which not only considers the model's learning ability of domain-invariant causal knowledge but also further considers the generalization ability of learned causal knowledge in unseen domains during training. By simulating the domain generalization scenario from visible source

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domains to unseen target domains, the model's generalization performance in domain generalization tasks is further improved.

G. Discussion

In this section, we have discussed the limitations of the proposed method for domain generalization from several perspectives: Temporal, spatial, spectral, and class types and numbers.

1) Discussion on Temporal and Spatial Shifts: CMG is adaptable to scenarios where the source and target domains are collected from different spatial regions or at different temporal. The experiments on the Houston dataset with temporal displacement and the GID dataset with spatial displacement validated the adaptability of CMG to the temporal and spatial shifts between the source and target domains.

2) Discussion on Spectral: CMG is adaptable to scenarios where there are spectral differences between the source and target HSI domains. In this section, we discuss CMG's ability to handle spectral differences between the source and target domains.

a) Difference in the Number of Spectral Bands: CMG requires the source and target domains to have the same number of spectral bands. However, when the source and target HSIs are collected from different sensors, they often have varying numbers of spectral bands. The experimental results on datasets with significant differences in spectral bands (e.g., HS13 and HS18 collected from different sensors) also demonstrate CMG's adaptability to such scenarios. In this case, it is necessary to unify the number of spectral bands in both domains to ensure consistency in input data. In this article, we employed band selection techniques to unify the spectral bands across the two domains. In fact, some dimensionality reduction techniques, such as principal component analysis (PCA) [46], minimum noise fraction (MNF) [47], or the dimension unification layer [48], can also serve as alternative solutions.

b) Difference in Spectral Distribution: CMG is capable of handling spectral distribution differences between the source and target domains. Due to variations in equipment, environment, and other factors during data collection, the spectral distributions of the source and target HSI domains often differ, which may hinder cross-domain knowledge transfer. In CMG, the B-VAE component decouples domain-specific distribution from the domain-invariant class distribution, reducing the impact of spectral distribution differences between domains. In addition, the dual-level optimization mechanism encourages the model to learn domain-invariant causal meta-knowledge from different domains, thus promoting cross-domain knowledge transfer. Experimental results on datasets with significant spectral distribution differences (e.g., HS13 and HS18 collected from different sensors, GIDWH and GIDDZ collected from different geographical regions) further validate CMG's adaptability to such scenarios.

3) Discussion on Class Types and Numbers: In HSI domain generalization, it is typically assumed that the class types and numbers are consistent between the source and target domains [19], [49], [50]. In this article, we also follow this common setting, ensuring that all datasets used have consistent class types and numbers, as shown in Tables I–III. The proposed method achieves the best classification performance under this setup. However, in real-world applications, it is more common for the source and target domains to have differences in class types and numbers, posing greater challenges for cross-domain knowledge transfer. In this section, we discuss CMG's adaptability when the class types and numbers differ between the source and target domains.

First, when the class types between the two domains are inconsistent, the knowledge learned in the source domain for specific classes may be difficult to apply to the target domain. In such cases, existing HSI domain generalization methods may not be suitable. For example, knowledge used to identify "Tree" in the source domain is not easily transferred to identifying "Water" in the target domain. Recent work has explored meta-learning-based solutions [11], [48], leveraging meta-learning's episodic training mechanism to learn cross-class shared knowledge, enabling cross-domain knowledge transfer when the class types between the source and target domains are inconsistent. However, it is important to note that these methods require a small number of labeled samples from the target domain during testing to compute class prototypes, making them difficult to directly apply to HSI domain generalization tasks with unlabeled target-domain samples.

Second, when the number of classes between the two domains is inconsistent, the source or target domain may have classes not present in the other domain. To accurately classify target-domain samples, the model must address the open-set recognition problem, which involves identifying both shared and private classes between the two domains during the inference phase. Existing HSI domain generalization methods based on closed-set assumptions are not suitable for this scenario. Although some researchers have explored domain generalization under open-set settings in natural image tasks, this area has yet to be investigated in HSI classification. In future work, we plan to explore the potential of CMR in open-set settings and study how to design more robust domain generalization frameworks that can handle class type and number inconsistencies, thus improving the model's adaptability to complex open-world scenarios.

4) Summary: CMG is capable of adapting to temporal, spatial, and spectral differences between the source and target domains, but it is only applicable in scenarios where the class types and numbers are consistent between the source and target domains. It struggles to handle open-set scenarios where the class types and numbers are inconsistent between the two domains. In addition, since the target domain is unseen during training in HSI domain generalization, the performance of CMG depends on the shareability of knowledge between the source and target domains. CMG tends to perform better when the source and target domains are geographically close (such as HS13–HS18) or when the data is collected using the same sensors (such as GIDWH-GIDDZ and PaviaU–PaviaC). However, when the source and target domains differ in both geographic regions and data collection equipment, the

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spectral properties of the same land-cover types may vary significantly. In such cases, due to the limited cross-domain shareable knowledge, CMG struggles to achieve satisfactory performance.

V. Conclusion

This article integrates decoupled learning, meta-learning, and causal learning into a unified framework for hyperspectral domain generalization, proposing CMG. CMG generates virtual domains by perturbing the domain distribution and constructs domain generalization meta-tasks based on these virtual domains to simulate the domain generalization scenario from visible source domains to unseen target domains. It then uses a DLOM to help the model induce domain-invariant causal meta-knowledge from these meta-tasks. CMG has the following advantages: 1) it can capture pure class identification information and domain-specific information from HSIs, achieving sufficient decoupling of domain distribution and class distribution, thereby reducing the marginal distribution differences between domains; 2) it can cut off the false statistical correlation between noncausal factors and labels, capturing the true causal relationship between data and labels even when noncausal factors are difficult to observe directly; 3) it can simulate domain generalization scenarios using only single-source domain, avoiding additional label costs and memory overhead. Experiments on two HSI datasets demonstrate that CMG achieves state-of-the-art performance in hyperspectral domain generalization tasks. It should be noted that CMG has significant potential for application in hyperspectral signal-processing tasks, including the recognition of crop pests and diseases as well as the identification of minerals. However, the proposed CMG is only applicable to closed-set scenarios where the class types in the source and target domains are consistent, lacking a mechanism to identify unknown classes. It is important to note that in the complex open-world environment, open-set scenarios where the class types and numbers differ between the source and target domains are more common. In such cases, the model needs to identify shared and private classes across the two domains during inference. In future work, we plan to explore the potential of CMG in open-set settings and investigate how to design a more robust domain generalization framework that can handle inconsistencies in class types and numbers across domains, enhancing the model's adaptability to a complex open world.

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